

Jaisal Sharma

Kathmandu, Nepal

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EDUCATION

Pulchowk Campus, Tribhuvan University, Bachelor in Electronics, Communication and Information Engineering Dec 2024
Lalitpur, Nepal

- **Relevant Coursework:** Artificial Intelligence, Data Structures and Algorithms, Computer Networks, Object Oriented Programming, Embedded Systems, Computer Architecture and Organization, Numerical Methods

RESEARCH INTERESTS

- Machine Learning and Deep Learning Systems
- NLP and Generative AI (LLM, RAG)
- AI security and vulnerabilities
- Trustworthy and Fairness in AI
- Computer Vision
- LLMs and Knowledge Graph Integration

RESEARCH EXPERIENCES

Research Intern (Remote, Informal) Nov 2024 – Apr 2025
With Dr. Lihui Liu, Assistant Professor - Wayne State University

Integrating Choice Differentiation and Knowledge Graph Reasoning for Enhanced LLM Multiple-Choice QA

- Participated in weekly meetings for research discussion, model design, and progress updates.
- Conducted literature review, model design, and experimental evaluations under faculty mentorship.
- Contributed to a Commonsense QA project integrating LLM-based reasoning with ConceptNet using GNNs.
- Independently designed and implemented a prototype that improved model accuracy from 25% to 50%+ on CSQA and OBQA benchmarks, despite using a smaller LM and limited computational resources.

Undergraduate Research Student April 2023 – May 2024
Department of Electronics and Computer Engineering, Pulchowk Campus, Nepal

Colorizing Grayscale Images using Generative Adversarial Networks(GANs): A Deep Learning Approach

- Conducted a comprehensive literature review on image colorization methods and deep learning frameworks.
- Proposed and implemented a generative adversarial network (GAN) for realistic image colorization.
- Researched and studied the colorization effectiveness in three different color models(LAB, YUV, and HSI)

TECHNICAL SKILLS

Programming Languages: C++, C, Python, SQL

Frameworks/Technologies: Pytorch, OpenCV, Langchain, Numpy, Pandas, Pytorch Geometric, Streamlit, Scikit-Learn, Flask, HTML, CSS

Tools: MS Office, Google Suites, Overleaf(Latex), MATLAB, Proteus Simulation

PUBLICATIONS

Peer Reviewed Conference Articles:

Automated Image Colorization using Generative Adversarial Networks Dec 2025

9th International Conference on Algorithms, Computing and Systems (ICACS), Bangkok, Thailand

Accepted for ACM Digital Library (Scopus & Ei Compendex) · **Paper received Best Presentation Award**

Jaisal Sharma, Rahul Shrestha, Dibakar R. Pant, Jukka-Pekka Skön, Jukka Heikkonen, Rajeev Kanth

PROJECTS

- GraLLM-Tutor: An Explainable AI Framework | LLM + Knowledge Graph**  Apr 2025
- Engineered a novel explainable AI (XAI) framework that synergizes a large language model (Llama 3) with a structured commonsense knowledge graph (ConceptNet) to ground AI-generated answers in verifiable logic.
 - Developed an automated reasoning pipeline that performs LLM-based concept extraction, queries for relational paths via the ConceptNet API, and synthesizes disparate knowledge fragments into a clear explanation.
- Medical Chatbot & Knowledge Retrieval Systems | RAG App**  June 2024
- Developed an AI-powered medical research assistant using LangChain and Pinecone, implementing advanced document retrieval techniques to process 5,699 text chunks from clinical literature
 - Engineered a hybrid NLP pipeline combining dense vector embeddings with LLM's generative capabilities
- SemDiff-QA: Semantic Differentiation for Multi-Choice Question Answering(MCQA)**  Nov 2025
- Implemented the **DCQA research paper (ECAI 2024)** by re-architecting a **T5-Base** model into a discriminative ranker, shifting from text generation to list-wise classification to resolve semantic drift.
 - Engineered a memory-efficient training pipeline using **Mixed Precision (FP16)** and gradient accumulation to optimize performance on T4 GPUs, achieving **60.52% accuracy** on the **CommonsenseQA** dataset.
- Defending Against RAG Prompt Injection Using DPO-Based Fine-Tuning**  Apr 2025
- Executed advanced **Indirect Prompt Injection** attacks on RAG systems using multi-turn poisoned knowledge bases and automatically generated adversarial queries.
 - Implemented a **Direct Preference Optimization (DPO)** defense with LoRA adapters, training the LLM to resist subtle non-imperative misinformation while preserving utility on benign queries.
- Image Colorization using Deep Learning(GANs) | Pytorch**  June 2023 – Feb 2024
- Proposed deep learning-based image colorization method utilizing generative adversarial networks (GANs)
 - Demonstrated efficacy in colorizing black and white images, offering a useful approach in computer vision applications.
- Graph Neural Networks Recommender System with LightGCN**  June 2024
- Developed a recommendation system leveraging LightGCN (Light Graph Convolutional Network) with supervised learning,
 - Fine-tuned supervised learning techniques to enhance model performance, optimizing hyperparameters and evaluating results using metrics such as precision and F1 score.
- Fine Tuning BERT-base-uncased Hugging Face Model on Kaggle Hate Speech Dataset | NLP**  Sep 2024
- Achieved improved accuracy in classifying toxic language, while applying techniques such as tokenization, model evaluation, and hyperparameter tuning to optimize performance for real-world NLP applications.

WORK EXPERIENCE

- Machine Learning Intern**, Real Time Solutions, *Lalitpur, Nepal* Jul 2024 - Oct 2024
- Adapted the RAFT (Recurrent All-Pairs Field Transforms) architecture using PyTorch to estimate river surface velocity, utilizing dense optical flow to quantify water motion from surveillance footage.
 - Applied this vision-based system for flood monitoring across Nepal's rivers, analyzing flow vectors to detect critical discharge rates and enhance non-intrusive early warning capabilities.

TEACHING EXPERIENCE

- Latex (Overleaf) Instructor**, IEEE Pulchowk Latex Workshop – Lalitpur, Nepal Jun 2023
- Conducted hands-on sessions guiding participants through LaTeX basics and advanced formatting techniques
 - Facilitated group discussions and provided personalized assistance to attendees during practical exercises to prepare lab reports, field reports and project reports.
- Software Fellowship Tutor**, Locus Pulchowk – Lalitpur, Nepal July 2023
- Delivered tutorial on basics of Python Programming and Data Structures to students at the campus.

GRANTS, SCHOLARSHIPS & AWARDS

Conference Participation Grant - Awarded by University of Turku, Finland	Oct 2025
Received a full financial support from the University of Turku for conference participation at ICACS 2025 Bangkok, Thailand	
Undergraduate Engineering Scholarship (All Nepal Rank 252 - Top 4%)	2019
Awarded a prestigious merit-based scholarship covering full tuition fees for the undergraduate degree by Institute of Engineering, Pulchowk Campus	
High School Excellence Award (GPA 3.75)	2017
Awarded a merit-based scholarship that provided significant fee waiver and facilities for high school degree by Nepal Government	

CERTIFICATIONS AND TRAINING

Licensed Electronics Engineer , Nepal Engineering Council	Oct 2024
AWS Academy Data Engineering , AWS Training and Certifications	Jun 2024
Python for Data Science, AI & Development , IBM Coursera	Mar 2024
Databases and SQL for Data Science with Python , IBM Coursera	Jul 2024
Machine Learning with Python , IBM Coursera	Jul 2024
Data Science Methodology , IBM Coursera	Jan 2024

LEADERSHIPS AND VOLUNTEERING

Volunteer , LOCUS IOE Pulchowk – Lalitpur, Nepal	2023, 2024
- Volunteered at Locus, a national technological festival held annually at Pulchowk Campus, where various software projects, robotics demonstrations, and drone environments are showcased. - Actively participated in organizing and supporting the drone competition during this event.	
Tutor , Children in Technology (CIT) Program 2024	Nov 2024
- Visited rural areas and various schools across Baglung District, where I delivered engaging presentations on digital transformation to students. - Provided presentations to school level students on the importance and benefits of integrating technology in education.	
Technical Coordinator , Pulchowk Music Club and Research Center	April 2023 – March 2024
- Organized and managed technical aspects of music fests, including sound engineering, equipment setup, and live performance coordination, ensuring seamless event execution. - Conducted workshops and other music-related events, providing technical training and fostering collaboration among participants, enhancing the club's overall engagement and impact.	